10 What is a statement of the principle of conservation of momentum?

A A force is equal to the rate of change of momentum of the object upon which it acts.

B In a perfectly elastic collision, the relative momentum of the objects before impact is equal to their relative momentum after impact.

C The momentum of an object is the product of the mass of the object and its velocity.

D The total momentum of a system of interacting objects remains constant, providing no resultant external force acts on the system.